

Supplementary Material: Graph-Constrained Reasoning with Bidirectional Sub-Answer Cross-Checking for Multi-Hop QA

Anonymous Submission

This document collects supporting analyses referenced in the main paper. Reviewers are not obligated to examine it; all core claims and evidence appear in the main text.

1 Qualitative Positioning

Table 1 clarifies the scope of GERS relative to representative reasoning paradigms; it is intended to clarify scope rather than claim full empirical dominance over every family.

Table 1: Qualitative positioning.

Method family	Structure	Execution	Checkable sub-ans.
CoT	linear text	implicit	no
CoT-SC	linear samples	implicit	no
ToT / GoT	tree or graph thoughts	search/flexible	state nodes
RwG / graph prompts	entity/context graph	prompt-cond.	usually no
MoDeGraph-style	dependency prompt	prompt-cond.	entity level
GERS	sub-question DAG	topological	yes, forward/backward

2 Consistency Score Bucket Diagnostics

Table 2 reports the CS-correctness breakdown for GERS-CV2 (HotpotQA, n=500). AUROC uses EM correctness as the binary label. Although higher CS buckets have higher EM accuracy, 156 of the 250 samples with CS=1.0 are still EM-wrong: internal agreement is not factual correctness.

Table 2: CS-correctness diagnostics (HotpotQA, n=500).

Score variant or bucket	Samples	EM	F1
Old structural CS AUROC	500	AUROC	0.498
New CS (GERS-CV) AUROC	500	AUROC	0.581
New CS (GERS-CV2) AUROC	500	AUROC	0.589
GERS-CV2, $CS < 0.5$	75	0.200	0.278
GERS-CV2, $0.5 \leq CS < 0.7$	86	0.244	0.333
GERS-CV2, $0.7 \leq CS < 0.9$	85	0.247	0.375
GERS-CV2, $CS = 1.0$	250	0.376	0.499

3 HotpotQA Per-Type EM

Table 3 shows that the gain concentrates on comparison/branch-merge samples; on bridge questions all methods are close because the context directly contains extractable evidence.

Table 3: HotpotQA per-type EM (n=500).

Method	bridge (n=404)	comparison (n=96)
Zero-Shot	0.218	0.521
CoT-SC	0.218	0.448
GERS-CV2	0.240	0.562

4 Computational Cost

Table 4 compares per-sample cost. GERS-CV2 (single-path) uses $\sim 6-8$ LLM calls and 6.6s latency—more calls than CoT-SC’s three samples, but comparable wall-clock time under the parallel API setting. The practical cost is moderated by adaptive routing, shorter verification prompts, and backward verification only on complex questions. The tested GERS-SC ($\sim 12-15$ calls) costs more with no performance gain under the original structural score.

Table 4: Cost comparison (HotpotQA, n=500, per-sample). LLM calls estimated from the pipeline (1 decomposition + n sub-steps + 1 aggregation, avg $n \approx 2$; GERS-CV2 adds n backward calls).

Method	LLM calls/q	Latency	EM	F1
Zero-Shot	1	0.7s	0.276	0.389
Std CoT	1	3.0s	0.264	0.368
CoT-SC (N=3)	3	8.3s	0.262	0.373
MoDeGraph-style	3	8.7s	0.252	0.366
GERS-Adap.	1-4	4.8s	0.284	0.395
GERS-SC (K=3)	$\sim 12-15$	16.2s	0.282	0.398
GERS-CV2	$\sim 6-8$	6.6s	0.302	0.413

5 Generalization and Additional Ablations

Second model (generalization). To check that the method is not specific to Qwen3-8B, we re-run on Qwen-Plus (a different, stronger model family) on 300 HotpotQA samples. GERS-CV2 (EM 0.363, F1 0.496) is on par with CoT-SC (EM 0.367, F1 0.494), diff -0.004 , $p = 1.000$. This indicates a generalization boundary: on a stronger base model, where Zero-Shot/CoT-SC already perform well, the structured method’s advantage disappears. The gain is concentrated on medium-capability models where explicit decomposition matters.

Weight design. Replacing the downstream-weighted w_i with uniform weights yields EM 0.300 / F1 0.411 (vs. 0.302 / 0.413), indistinguishable. The weighting design does not affect end performance; it is kept for interpretability.

2Wiki at larger scale. Extending 2WikiMultiHopQA to 300 samples confirms the boundary: GERS-CV2 EM 0.390 / F1 0.462 remains below Standard CoT EM 0.443 / F1 0.521, with the gap concentrated on bridge_comparison. The larger sample makes the diagnostic boundary more reliable.

Evidence-grounded verification (diagnostic). We test an evidence-grounded variant on the first 100 samples, where each backward sub-answer must quote a supporting context span and

the score multiplies an evidence-grounding heuristic. On HotpotQA it preserves EM and slightly changes F1 (GERS-CV2 0.451 vs. grounded 0.455) while lowering CS from 0.781 to 0.730. On 2Wiki it preserves EM and stays close in F1 (0.469 vs. 0.463), and improves bridge_comparison F1 from 0.524 to 0.571 within the same diagnostic rerun, but loses on other types. Because this was a separate diagnostic rerun, its per-type numbers should be compared only against the same-run GERS-CV2 row. Evidence grounding suppresses inflated self-consistency scores and helps some bridge cases, but is not yet a robust end-task improvement.

Repair variants (negative result). Repair variants regenerate low-confidence sub-answers. On HotpotQA first-100, `gers_repair` reaches EM 0.300 / F1 0.401 / CS 0.847 (51 repair triggers) and `gers_repair_soft` EM 0.320 / F1 0.411 / CS 0.874 (47 triggers), versus GERS-CV2 EM 0.330 / F1 0.451 / CS 0.781 (0 triggers). On 2Wiki first-100, `gers_repair_soft` EM 0.340 / F1 0.403 / CS 0.851 (53 triggers) vs. GERS-CV2 EM 0.400 / F1 0.469 / CS 0.745. Repair raises CS while reducing EM/F1, so it remains a negative result and is not a main method.

6 Implementation and Reproducibility

Model and API. Qwen3-8B via DashScope (OpenAI-compatible) with `enable_thinking=False`; Qwen-Plus for the generalization run. **Parallelism:** `ThreadPoolExecutor` with 4 workers (4 is the stable sweet spot; 6+ workers trigger DashScope QPM 429, handled by exponential-backoff retry at 1/2/4s on `RateLimitError`/`APITimeoutError`/`APIConnectionError`/`InternalServerError`; `max_retries=0`, `timeout=60`). **Seeds:** bootstrap uses `numpy.random.default_rng(42)` with $B = 10000$; the result is stable across seeds 0/1/42/123/2024. **Statistics:** paired bootstrap resamples per-sample differences (not unpaired); McNemar uses chi-square with Edwards' continuity correction. Reproducer: `experiments/_paired_stats.py`. **Datasets:** HotpotQA and 2WikiMultiHopQA are publicly available with the cited references. **Code:** the implementation, experiment runner, figure/Word generators, and the stats reproducer are provided in the code archive accompanying this submission.